

salt. It is just an indication of what is to come, and not a clear road-map. With that in mind, we have been through the technology trends predicted by major research firms and analysts for 2024 and jotted down the key points for you to mull over. As the year goes by, we will keep updating you on what really transpires.

Another year hijacked by generative AI

GenAI's growth spree. GenAI and large-language models (LLMs) will continue to be big. It is OpenAI's generative pre-training transformer (GPT) model and its chatbot ChatGPT that kind of commoditised AI for anyone and everyone to use.

While ChatGPT has become a household name now, there are others worth their salt like OpenAI Dall-E, Anthropic Claude, Google Bard, Duet AI, Github Copilot, Cohere Generate, and so on. We are also bound to see OpenAI's GPT Store gain popularity, and similar AI app stores coming up from other players. GenAI is also expected to influence the search industry, bringing about a paradigm shift.

While AI already works behind the scenes, we are bound to see more search engines incorporating more user-facing AI features. GenAI trendwatchers suggest that modalities like image and audio will become more impactful than text—such as the creation of molecular structures, visual AI models, interactive voice response (IVR) in local languages, and so on.

AI-assisted development. We are bound to see genAI and ML aiding software engineers in the design, development, and testing of applications. This will improve the productivity of software engineers by reducing the time spent on writing code and making time available for strategic activities like design.

While there is the fear of AI robbing people of their jobs, it is more likely that AI will support developers rather than replace them.

GenAI features in software.

Experts feel there is no doubt that genAI features will be included in several software platforms this year. But the real challenge will lie in gauging whether users are willing to pay for these features, and how much. Microsoft, for example, is launching Copilot for Microsoft 365, with a monthly subscription fee.

AI in enterprises, small and large

Democratisation of AI. Gartner projects that by 2030, every dollar of IT spending will have an AI component. Gartner predicts that the democratisation of AI is going to gain steam this year. The report predicts that by 2026, over 80% of enterprises will have used generative AI application programming interfaces (APIs) and models, and/or deployed generative AI-enabled applications in production environments, up from less than 5% in early 2023.

Enterprises likely to train

AI with private data. AI models trained on public data are unreliable in many aspects. According to a Deloitte report, "Because generative models have required such massive volumes of training data, the first wave of public models were mainly trained on the largest data set available: the public internet. Thus, these models have also absorbed the many biases, contradictions, inaccuracies, and uncertainties of the internet itself... When pressed for facts, models trained on public data, such as social network posts, may fabricate them. And they can do so with authority, causing many users to believe their assertions without properly fact-checking the results. Popular LLMs were not designed to be factually accurate, but

rather statistically accurate."

To avoid risks of misinformation and hallucination, Deloitte predicts that enterprises might turn to training their AI models using their private enterprise data collected over several years. This will enhance productivity and optimise costs while unlocking complex insights. This trend is likely to further the demand for MLOps partners.

AI TRISM. As you are likely to have read in our previous issue, world leaders got together at the AI Safety Summit held in the UK in November 2023 to discuss how to put in proper guardrails to mitigate the risks involved in AI. As India's Minister of State for Electronics and IT, Rajeev Chandrasekhar, rightly said at the summit, AI should not be demonised to the extent that it is regulated out of existence. It is a kinetic enabler of India's digital economy and presents a big opportunity for us. At the same time, he acknowledged that proper regulations must be in place to avoid misuse of the technology. He opined that in the past decade, countries across the world, including ours, inadvertently let regulations fall behind innovation and are now having to contend with the menace of toxicity and misinformation across social media platforms. As AI has the potential to amplify toxicity and weaponisation to the next level, he said that countries should work together to be ahead, or at least at par with innovation when it comes to regulating AI.

One potential solution that came up at the summit sought tech companies to share their latest AI models with government agencies who will test it comprehensively with advanced computing systems to ensure safety and minimise risks before public release. Even as those high-level efforts are set into motion, it is important that organisational